

Business Background

A leading retail company wants to better understand its customers' shopping behavior. The primary goal of this analytics initiative is to improve sales metrics, optimize customer satisfaction, and build long-term loyalty.

Problem Identification

The management team has noticed shifts and changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). Currently, they lack clarity on which specific factors—such as applied discounts, product reviews, seasonal trends, or payment preferences—most strongly drive consumer decisions and repeat purchases.

Project Objective

The main task of this project is to dissect and analyze the company's "Consumer Behavior" dataset to extract actionable insights.

Overarching Business Question

All data analysis efforts (from Python exploration to SQL queries and Power BI dashboards) are focused on answering this single strategic question:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Deliverables

1. **Data Preparation & Modeling (Python):** Clean and transform the raw dataset for analysis.
2. **Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. **Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. **Report and Presentation:** Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
5. **GitHub Repository:** Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.